

5G-LENA **NS-3**

5G-LENA with Network Simulator 3

Install & Use

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01 OVERVIEW OF 5G-LENA

OVERVIEW

➔ **5G-LENA** (5G Link-Level and End-to-End Network Simulator) is an extension module for NS-3 developed by **CTTC** (Centre Tecnològic de Telecomunicacions de Catalunya)



OBJECTIVES

➔ Comprehensive simulation of the 5G New Radio (NR) network system from the physical (PHY) layer to the network layer.

➔ Supports both link-level and end-to-end simulations.

➔ Serving **research purposes:**

- 5G wireless access
- Radio Resource Management (RRM)
- Beamforming, MIMO
- Core network integration

01

OVERVIEW OF 5G-LENA

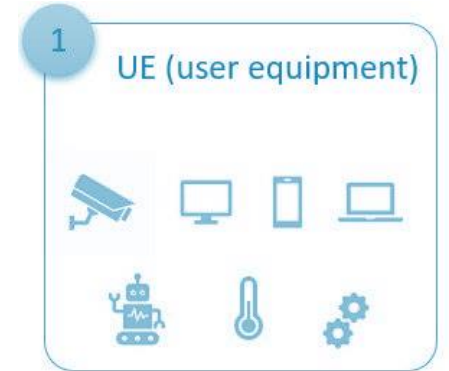
FEATURES

FEATURES	DESCRIBE
NSA Architecture (Non_Standalone)	Based on LTE EPC to simulate 5G connectivity integrated with the 4G network.
3GPP TR 38.901 Channel Model	Accurate 3D radio channel model for the frequency range from 0.5 to 100 GHz
TDD/FDD Support	Flexible support for Time/Frequency Division Duplexing configurations
MIMO & Beamforming	Supports dynamic beamforming, precoding, and antenna array modeling
QoS & Scheduler	Scheduling algorithms: RR, PF, ...

01 OVERVIEW OF 5G-LENA

COMPONENTS OF THE SIMULATION

- ➔ **gNB (base station):** Configurable number, positions, etc.
- ➔ **UE (User equipment):** Simulate user equipment with mobility and measurement capabilities
- ➔ **NR PHY/MAC/RLC/PDCP:** Main protocol layers of 5G NR
- ➔ **EPC Integration:** Connection to LTE core network to simulate NSA architecture



02 INSTALL PROCESS ON NS3

Step 1: Open the Terminal on Ubuntu and install the prerequisite packages for NR:

```
sudo apt-get install lbc6-dev  
sudo apt-get install sqlite sqlite3 libsqlite3-dev
```

Step 2: Download NS3

```
git clone https://gitlab.com/nsnam/ns-3-dev.git  
Cd ns-3-dev
```

Step 3: Download module NR

```
cd contrib  
git clone https://gitlab.com/cttc-lena/nr.git
```

Step 4: Checkout the latest NR release branch

```
cd nr  
git checkout 5g-lena-v2.5.y
```

Step 5: Checkout the newest NS-3 release branch.

```
cd ../../  
git checkout ns-3.38
```

Step 6: build NS3+ NR

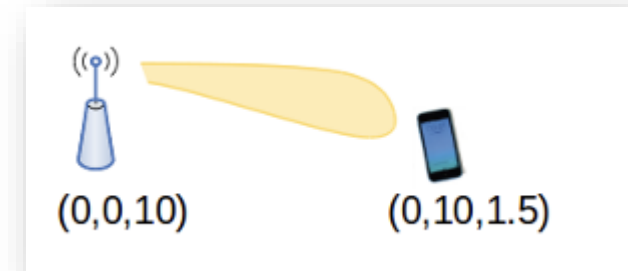
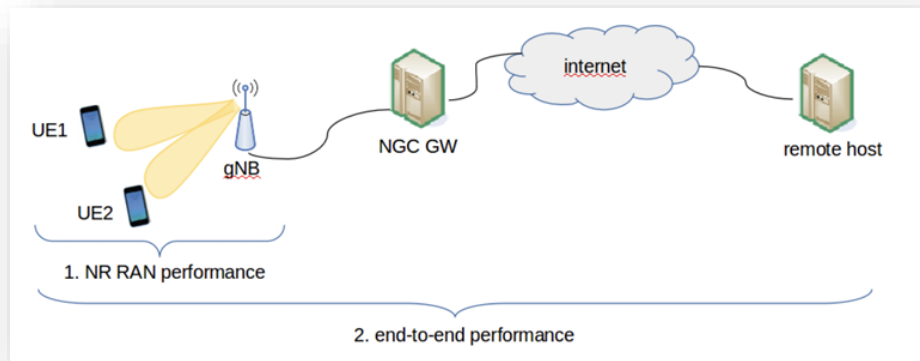
```
./ns3 configure --enable-examples --enable-tests  
./ns3 build
```

03

DEMO MỘT SỐ KỊCH BẢN ĐƠN GIẢN

➔ Demonstrate two simple scenarios available in the '**examples**' folder within the main working directory **ns-3-dev**.

■ **cttc-3gpp-channel-simple-ran**



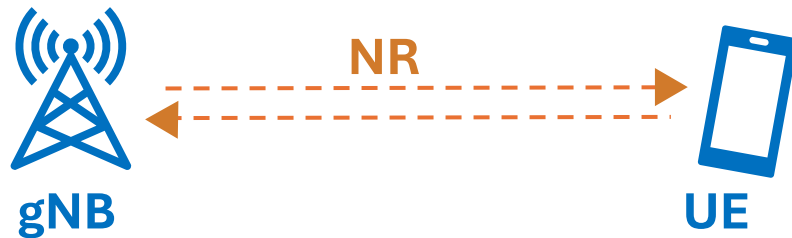
■ **cttc-3gpp-channel-nums**

03

DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Network Simulation



➡ The network architecture includes **one gNB**
and one UE

➡ PointToPoint model using NR (New Radio – 5G wireless technology)

➡ Frequency: **28 GHz** , Bandwidth: **400 MHz**

➡ Send a **UDP packet** of **1000 bytes** at **0.4 seconds** after the simulation starts.

➡ Protocol Stack:

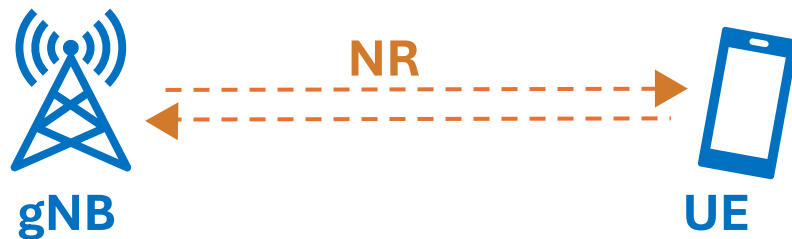
- **PDCCP** (Packet Data Convergence Protocol)
- **RLC** (Radio Link Control)
- **MAC/PHY**

03

DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Scenario objective



➡ Log the pdcpDelay value upon packet arrival at the PDCP layer.

➡ Log the information when the packet arrives at the RLC layer:

- **RNTI** (Radio Network Temporary Identifier)
- **LCID** (Logical Channel ID)
- **RLC delay** (rlcDelay)

03

DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Scenario objective

➡ **RNTI** (Radio Network Temporary Identifier) → A temporary identifier assigned to a UE by the network when it connects to a gNB (5G base station).

- Used to identify the UE at the PHY and MAC layers (i.e., to determine which UE a data packet is destined for)
- Each RNTI is unique within the coverage area of a single gNB

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DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Scenario objective

- **LCID (Logical Channel ID)** → A logical identifier used to distinguish between different logical channels operating under the same RNTI (i.e., on the same UE).
- Each UE can have multiple types of logical channels (from LCID 1 to 10+) for various purposes:
 - **Control plane traffic (SRB):** Signaling Radio Bearers (SRB1, SRB2,...)
 - **User plane traffic (DRB):** Data Radio Bearers (DRB1, DRB2,...)

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DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Scenario objective

➔ **RLC** (Radio Link Control) ➔ is the intermediate layer between PDCP and MAC

- **Segmentation/ Concatenation**
- **ARQ (Automatic Repeat Request)**
- **Buffering**
- **Reordering**

➔ **Latency depend on:**

- Size of buffer
- MAC processing speed
- Polling interval and status reporting
- ACK window size (AM mode)
- Packet loss and number of retransmissions

03

DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Run Demo

Step 1: Open the Terminal and navigate to the main working directory ns-3-dev using the following command: **cd ns-3-dev.**

Step 2: Type this command to run simulation
./ns3 run cttc-3gpp-channel-simple-ran

```
kuyen@kuyen-VirtualBox:~$ cd ns-3-dev
kuyen@kuyen-VirtualBox:~/ns-3-dev$
```

```
kuyen@kuyen-VirtualBox:~/ns-3-dev$ ./ns3 run cttc-3gpp-channel-simple-ran
[0/2] Re-checking globbed directories...
ninja: no work to do.

Sending data in downlink.

Data received at RLC layer at:+4.02243e+08ns

rnti:1

lcid:3

bytes :1024

delay :2242855

Packet PDCP delay:2242855
kuyen@kuyen-VirtualBox:~/ns-3-dev$
```

03

DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Run Demo

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Packet PDCP delay:2242855
kuyen@kuyen-VirtualBox:~/ns-3-dev$
```

The simulation is sending data from gNB to UE.

→ This data passes through the protocols: **PDCP**

→ **RLC → MAC → PHY → UE.**

RLC received data from MAC/PHY at
402.234 ms

1 is the ID assigned to the UE receiving data in the simulation → the gNB uses this value to identify the corresponding UE at the MAC and PHY layers.

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DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Run Demo

```
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delay :2242855
Packet PDCP delay:2242855
kuyen@kuyen-VirtualBox:~/ns-3-dev$
```

3 typically corresponds to **DRB1** – the bearer designated for user data → **user plane traffic**

The size of the received data is **1024 bytes**

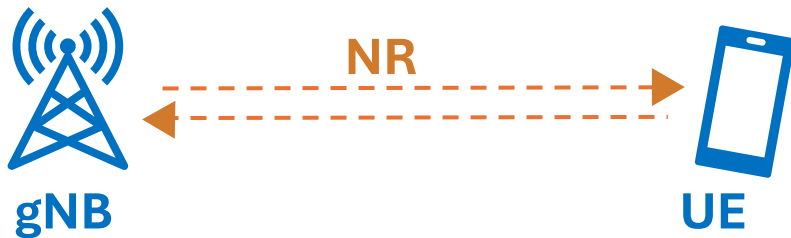
Packet delay from **PDCP** at gNB to **RLC** at UE (unit: ns)

03

DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-nums

■ Network Scenarios



➔ The network architecture includes one gNB and one UE

➔ PointToPoint model using NR (New Radio – 5G wireless technology)

➔ Frequency: **7 GHz** (mid-band), Bandwidth: **100 MHz**

➔ Send a **UDP packet** of **1000 bytes** at **0.4 seconds** after the simulation starts.

➔ Use Flow Monitor to:

■ Theo dõi các thông số của mô hình mạng
(throughput, delay, jitter, packet loss)

■ Calculate the average metrics
(Mean throughput, mean delay,...)

03

DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Run Demo

Step 1: Open the Terminal and navigate to the main working directory `ns-3-dev` using the following command: **`cd ns-3-dev`**.

```
kuyen@kuyen-VirtualBox:~$ cd ns-3-dev
kuyen@kuyen-VirtualBox:~/ns-3-dev$
```

Step 2: Type this command to run simulation

`./ns3 run cttc-3gpp-channel-nums`

```
kuyen@kuyen-VirtualBox:~/ns-3-dev$ ./ns3 run cttc-3gpp-channel-nums
[0/2] Re-checking globbed directories...
ninja: no work to do.
Flow 1 (1.0.0.2:49153 -> 7.0.0.2:1234) proto UDP
  Tx Packets: 28126
  Tx Bytes: 28913528
  TxOffered: 385.513707 Mbps
  Rx Bytes: 28720264
  Throughput: 383.557857 Mbps
  Mean delay: 3.533664 ms
  Mean jitter: 0.041721 ms
  Rx Packets: 27938

  Mean flow throughput: 383.557857
  Mean flow delay: 3.533664

  UDP throughput (bps) for UE with node ID 0:372.506667
kuyen@kuyen-VirtualBox:~/ns-3-dev$
```

03

DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Run Demo

```
kuyen@kuyen-VirtualBox:~/ns-3-dev$ ./ns3 run cttc-3gpp-channel-nums
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Mean flow throughput: 383.557857
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UDP throughput (bps) for UE with node ID 0:372.506667
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```

UDP data flow between the Client (UE): 1.0.0.2 port 49153 and the Server: 7.0.0.2 port 1234.

Number of UDP packets sent

Theoretical throughput attempted to be sent: **385.51 Mbps.**

Actual throughput received: **383.55 Mbps.**

Mean delay

Low variation in delay between consecutive packets indicates a stable network

03

DEMO SIMPLE SCENARIOS

cttc-3gpp-channel-simple-ran

■ Run Demo

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UDP throughput (bps) for UE with node ID 0:372.506667
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```

Number of packets received → packet loss

Repeat the average throughput value of the main flow.

The individual traffic measurement for the UE with node ID = 0 is lower (~**372 Mbps**) compared to **383 Mbps**, possibly due to minor packet loss.

THANK YOU